Introduction to Information Retrieval

Lecture 1: The IR Process, Vector Space Model & Boolean Retrieval

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Information Retrieval

February 9, 2026

Course Overview

1. **Introduction to IR: Vector Space Model & Boolean Retrieval** ← Today
2. Indexing: Sparse and Dense
3. Query Understanding
4. Ranking: BM25, Query Expansion, Language Models
5. Evaluation of IR Systems
6. Embeddings & Transformer-based Re-rankers
7. Dense Retrieval
8. Learning to Rank
9. Recommender Systems
10. Retrieval Augmented Generation (RAG)
11. LLMs as a Judge

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Today's Goals:

- Understand what IR is and why it matters
- Learn the fundamental IR process

Outline

1. What is Information Retrieval?
2. Challenges in IR
3. The Information Retrieval Process
4. Boolean Retrieval
5. The Vector Space Model
6. Looking Ahead
7. Summary and Assignment

What is Information Retrieval?

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The Essence of IR: Given an Information Need

- **Find relevant results** that satisfy the information need
- **By searching over a large collection** of items (billions of documents)
- **In reasonable time:**
  - Web search: < 1 second
  - AI/conversational systems: allow more time for generation
- **Present results** in a manner that is easily consumable by humans

IR vs. Database Systems

Aspect	Database Systems	IR Systems
Data	Structured (tables, schemas)	Unstructured (text, documents)
Queries	Precise (SQL)	Vague (natural language)
Matching	Exact	Approximate/Ranked
Result	All matching records	Top-k relevant documents
Failure	No results or all results	Ranked list (best effort)
Example	`SELECT * WHERE age>30`	"climate change effects"

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Key Difference

Databases: *"Give me all records matching X"*
IR: *"Give me the most relevant documents about X"*

IR vs. Natural Language Processing (NLP)

The Relationship

IR and NLP are closely related but have different focal points:

NLP (Language Centric)

- Focus on language **understanding**, representation, and **generation**.
- Deep analysis of syntax, semantics, and pragmatics.
- Models like BERT, Llama, GPT.

IR (Scale & Need Centric)

- Focus on **information seeking** at a massive scale.
- Uses NLP tech as a toolkit to model relevance.
- Priority: Speed, ranking, and satisfying user needs.

Typical Tasks

- **NLP:** Named Entity Recognition, Dependency Parsing, Translation, Summarization.
- **IR:** Web Search, Recommendation Systems, Ad-hoc Retrieval, Question Answering.

Real-World IR Applications

Web Search

- Google, Bing, DuckDuckGo
- Billions of web pages
- Sub-second response time

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- Amazon product search
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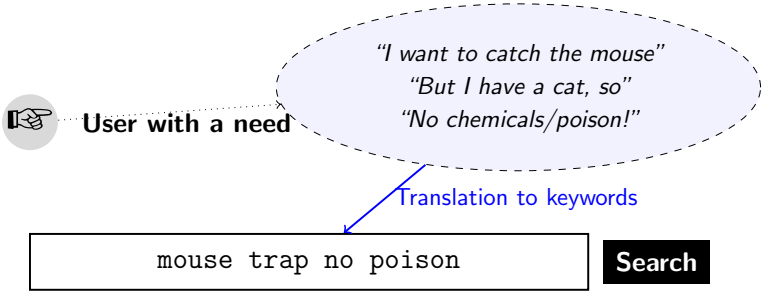
Specialized Domains

- Legal document search (LexisNexis)
- Academic papers (Google Scholar)
- Medical records (PubMed)
- News archives

IR is everywhere you search for information!

Challenges in IR

The Retrieval Challenge: Understanding Intent



The Semantic Gap

- **Syntactic match:** Docs containing "mouse", "trap", "poison".
- **User intent:** Docs describing *humane* traps or *mechanical* traps.
- **Danger:** Returning docs about "how poison traps work" (contains keywords!).

The Scale of Modern IR

Web Search Statistics (2024)

- **Google:** Processes 8.5+ billion searches per day
- **Index size:** 100+ trillion web pages indexed
- **Freshness:** New pages indexed within hours
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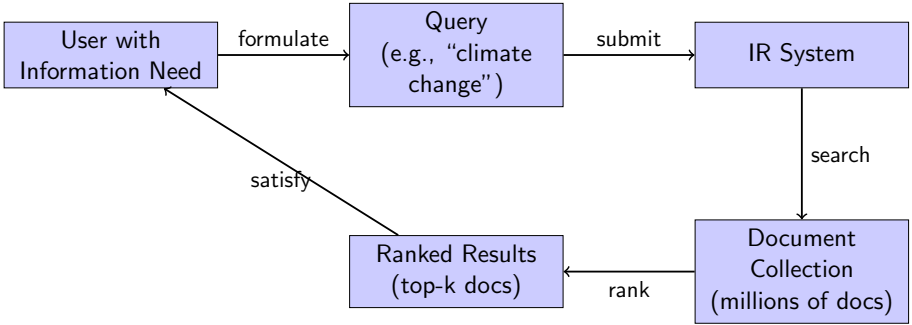
Challenges at Scale

- **Storage:** How to store 100 trillion documents?
- **Speed:** How to search in milliseconds?
- **Relevance:** How to rank billions of results?
- **Freshness:** How to keep index up-to-date?
- **Quality:** How to fight spam and misinformation?

The Information Retrieval Problem



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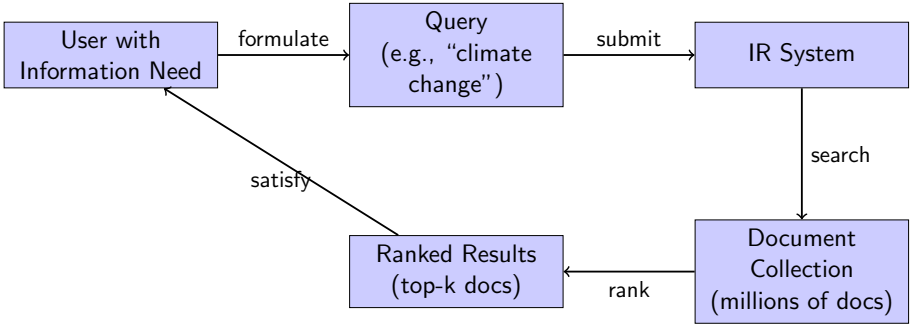

Core Questions:

1. How to represent documents and queries?
2. How to match queries to documents efficiently?
3. How to rank documents by relevance?

The Information Retrieval Process

Forward Reference: Lecture 2

The fundamental IR process, including pipeline architectures and preprocessing, is now covered in detail in **Lecture 2: Indexing**.

Why the change?

- Better logical flow: Preprocessing is tightly coupled with building indexes.
- Focus for today: Understanding the models (Boolean and Vector Space).

Proceeding to Section 3: Boolean Retrieval...

Boolean Retrieval

Boolean Retrieval Model

Definition

Documents are retrieved if they satisfy a Boolean expression of query terms

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Boolean Operators:

- **AND:** Both terms must appear (intersection)
- **OR:** Either term must appear (union)
- **NOT:** Term must not appear (negation)

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Example Queries

- `climate AND change` → docs containing both "climate" and "change"
- `cat OR dog` → docs containing "cat" or "dog" or both
- `java NOT coffee` → docs with "java" but not "coffee"
- `(climate AND change) OR (global AND warming)`

Boolean Retrieval with Sets

Documents as Sets of Terms

Document Collection:

- D1: {cat, sat, mat}
- D2: {dog, sat, log}
- D3: {cat, dog, run}
- D4: {bird, fly, sky}

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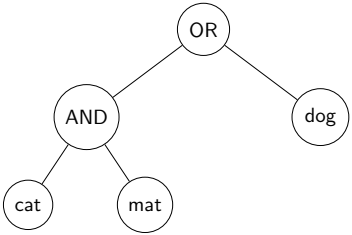
Query Processing:

- `cat AND dog`: {D1, D3} $\cap$ {D2, D3} = {D3}
- `cat OR dog`: {D1, D3} $\cup$ {D2, D3} = {D1, D2, D3}
- `cat NOT dog`: {D1, D3} - {D2, D3} = {D1}

Boolean Retrieval Example

Query: `(cat AND mat) OR dog`



Execution:

1. `cat`: {D1, D3}
2. `mat`: {D1}
3. `cat AND mat`: {D1, D3} $\cap$ {D1} = {D1}
4. `dog`: {D2, D3}
5. `(cat AND mat) OR dog`: {D1} $\cup$ {D2, D3} = {D1, D2, D3}

Advantages of Boolean Retrieval

✓ Advantages

- ✓ **Precise control:** Users specify exactly what they want
- ✓ **Transparent:** Clear why documents matched
- ✓ **Efficient:** Fast set operations on posting lists
- ✓ **Predictable:** Same query always returns same results
- ✓ **Expressive:** Can construct complex queries

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- Legal document search (very precise requirements)
- Patent search (need exact specifications)
- Database-like queries on text
- Expert users who know what they want

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✗ **Result set size unpredictable**

- Could return 0 results or 10 million
- Users want "top 10", not "all matching"

Boolean Retrieval: The Feast or Famine Problem

Too Restrictive (Famine)

- Query: `machine AND learning AND neural AND networks`
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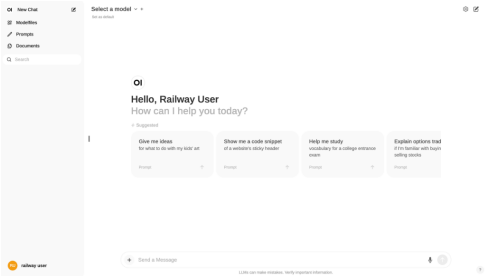
This motivates the Vector Space Model!

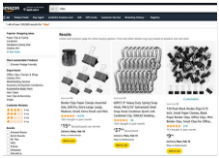
Examples of Search Systems

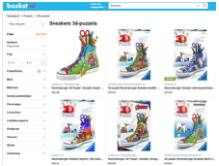
Conversational AI Search



ChatGPT - AI-powered conversational search

E-Commerce Search





Amazon - Product search and recommendations

Common Thread

All search systems share the same core challenge: finding relevant items from massive collections in real-time.

The Vector Space Model

From Boolean to Vector Space

Key Insight

Instead of documents being sets of terms (in/out), represent them as **vectors** with weights for each term

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- Weighted: importance/frequency

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- **Ranked:** D1, then D2

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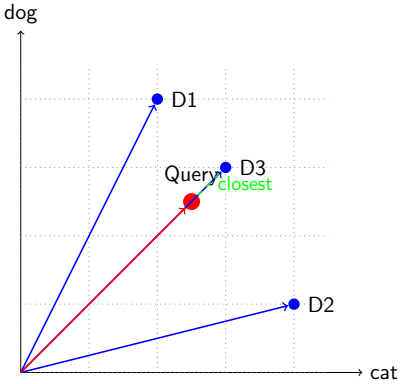
Advantage

D2 still returned (has "cat"), just ranked lower than D1

Vector Space Model: Geometric View

Documents and queries as vectors in term space



Similarity = Angle between vectors
(Smaller angle = more similar)

Term Weighting in Vector Space Model

Question: How to assign weights to terms?

Option 1: Binary Weights

$$w_{t,d} = \begin{cases} 1 & \text{if term } t \text{ appears in doc } d \\ 0 & \text{otherwise} \end{cases}$$

- Simple but ignores frequency

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- Problem: Common words get high weight

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TF-IDF Weighting

Idea: Balance term frequency with term rarity

TF-IDF Formula

$$w_{t,d} = \text{tf}(t, d) \times \text{idf}(t)$$

where:

- $\text{tf}(t, d) = \text{count}(t, d)$ (term frequency in document)
- $\text{idf}(t) = \log \frac{N}{\text{df}(t)}$ (inverse document frequency)
- N = total number of documents
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Intuition:

- **High TF:** Term appears often in this document $\rightarrow$ important for this doc
- **High IDF:** Term is rare across collection $\rightarrow$ discriminative
- **High TF-IDF:** Important AND discriminative

TF-IDF Example

Collection: 10,000 documents

Term	tf(t,d)	df(t)	idf(t)	tf-idf
neural	5	100	$\log(10000/100) = 2.0$	10.0
network	3	500	$\log(10000/500) = 1.3$	3.9
the	50	9000	$\log(10000/9000) = 0.05$	2.5

Observations:

- "neural" appears less than "the" (5 vs. 50) but gets higher weight (10.0 vs. 2.5)
- Rare, discriminative terms weighted higher
- Common words automatically downweighted

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Document Vectors with TF-IDF

Example Collection:

- D1: "cat cat dog"
- D2: "cat mat"
- D3: "dog bird bird"

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Vocabulary: {cat, dog, mat, bird}

Doc	cat	dog	mat	bird
D1	$2 \times \log(3/2) = 0.35$	$1 \times \log(3/2) = 0.18$	0	0
D2	$1 \times \log(3/2) = 0.18$	0	$1 \times \log(3/1) = 0.48$	0
D3	0	$1 \times \log(3/2) = 0.18$	0	$2 \times \log(3/1) = 0.95$

Vector Representation:

- D1: [0.35, 0.18, 0, 0]
- D2: [0.18, 0, 0.48, 0]
- D3: [0, 0.18, 0, 0.95]

Cosine Similarity

How to measure similarity between vectors?

Cosine Similarity Formula

$$\text{sim}(\vec{q}, \vec{d}) = \frac{\vec{q} \cdot \vec{d}}{|\vec{q}| \times |\vec{d}|} = \frac{\sum_t q_t \times d_t}{\sqrt{\sum_t q_t^2} \times \sqrt{\sum_t d_t^2}}$$

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Properties:

- Range: [-1, 1] (or [0, 1] for positive weights)
- 1 = identical vectors (same direction)
- 0 = orthogonal (no common terms)
- -1 = opposite vectors (not relevant for IR)

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Why cosine instead of dot product?

- Cosine normalizes for document length
- Long documents don't automatically score higher

Cosine Similarity: Example Calculation

Query: "cat dog" $\rightarrow \vec{q} = [0.5, 0.5, 0, 0]$
Documents:

- D1: [0.35, 0.18, 0, 0]
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Similarity Calculations:

$$\begin{aligned}
\text{sim}(q, D1) &= \frac{(0.5 \times 0.35) + (0.5 \times 0.18)}{\sqrt{0.5^2 + 0.5^2} \times \sqrt{0.35^2 + 0.18^2}} \\
&= \frac{0.265}{0.707 \times 0.394} = 0.95 \\
\text{sim}(q, D2) &= \frac{0.5 \times 0.18}{0.707 \times 0.511} = 0.25 \\
\text{sim}(q, D3) &= \frac{0.5 \times 0.18}{0.707 \times 0.967} = 0.13
\end{aligned}$$

Ranking: D1 (0.95) ¿ D2 (0.25) ¿ D3 (0.13)

VSM vs. Boolean Retrieval

Aspect	Boolean	Vector Space
Matching	Exact (in/out)	Partial (weighted)
Ranking	No	Yes (by similarity)
Result size	Unpredictable	Top-k
Term weighting	Binary	TF-IDF
Query expression	Boolean operators	Term list
Robustness	Brittle	Graceful degradation
User friendliness	Expert users	General users
Relevance	All equal	Ranked by relevance

Modern Usage

- **Boolean:** Still used in legal, patent search (precision critical)
- **Vector Space:** Foundation for web search, most IR systems
- **Hybrid:** Many systems combine both approaches

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✓Key Advantages

1. **Ranking by relevance**
   - Documents ordered by similarity to query
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Preview of Course

1. **Lecture 2 (Indexing):** How to build inverted indexes efficiently at scale
2. **Lecture 3 (Ranking):** Better weighting (BM25), query expansion, language models
3. **Lecture 4 (Transformers):** Neural re-ranking with BERT
4. **Lecture 5 (Dense Retrieval):** Semantic matching with embeddings
5. **Lecture 6 (Learning to Rank):** Learning optimal ranking from data

Preview: Lecture 2 — Indexing

Why Indexing Matters

- Building inverted indexes for billions of documents
- Sparse vs. dense representations
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Connection: Today we used inverted indexes conceptually; next lecture we'll build them efficiently

Preview: Lecture 3 — Ranking Models

Beyond TF-IDF

- **BM25:** Probabilistic ranking with saturation
- **Language models:** Generative approach to ranking
- **Query expansion:** Automatic term expansion with RM3
- **Pseudo relevance feedback:** Learn from top results

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- How to handle term saturation and length normalization
- How to expand queries automatically
- How to implement these in PyTerrier

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Connection: TF-IDF is foundation; BM25 and LMs refine it with better theory and empirics

Preview: Lectures 4-6

Lecture 4: Transformer Re-rankers

- BERT and transformer architectures
- Contextualized word embeddings
- Re-ranking with neural models
- "bank" (river) vs. "bank" (financial) finally distinguished!

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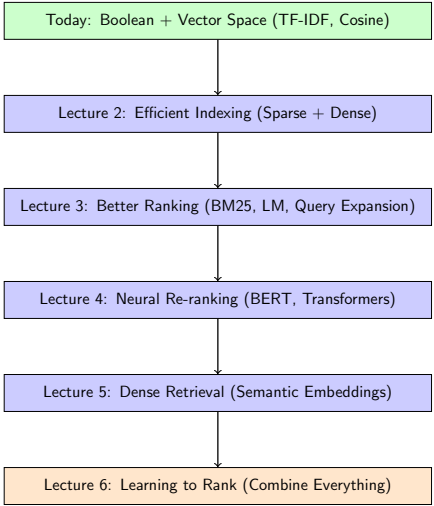
Lecture 6: Learning to Rank

- Combine features from all models
- Learn optimal ranking from labeled data
- LambdaMART, RankNet, etc.
- Put it all together

The Modern IR Stack



Summary and Assignment

Key Takeaways

1. Information Retrieval Fundamentals

- IR: Finding relevant unstructured documents at scale
- Different from database systems (approximate, ranked)
- Core challenges: representation, matching, ranking

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3. Boolean Retrieval

- Exact matching with AND/OR/NOT operators
- Fast, transparent, but no ranking
- Feast or famine problem

Key Takeaways (continued)

4. Vector Space Model

- Documents and queries as weighted term vectors
- TF-IDF weighting: term frequency $\times$ inverse document frequency
- Cosine similarity for ranking
- Solves Boolean's limitations: ranked results, partial matching

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VSM is foundation for modern IR, but has limitations (bag of words, no semantics) that we'll address in future lectures

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Assignment 1: Build Your First Search Engine

Due: Before Lecture 2

Part 1: Text Preprocessing (20 points)

- Implement tokenization, lowercasing, stop word removal
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Part 2: Boolean Retrieval (30 points)

- Build inverted index from preprocessed documents
- Implement AND, OR, NOT query processing
- Test on 10 queries, show results
- Analyze: when does Boolean work well/poorly?

Assignment 1 (continued)

Part 3: Vector Space Model (40 points)

- Implement TF-IDF weighting
- Implement cosine similarity
- Build ranked retrieval system
- Compare with Boolean on same 10 queries
- Report precision@10, recall for both systems

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Part 4: Analysis (10 points)

- Show examples where VSM outperforms Boolean
- Show examples where Boolean might be preferred
- Discuss trade-offs: precision vs. recall
- Identify limitations of your implementation

Deliverable: Code + report (3-5 pages) + demo on sample queries

Starter Code and Dataset

Provided Materials

- **Dataset:** TREC Disks 1-2 subset (10,000 documents) OR Cranfield collection
- **Queries:** 50 test queries with relevance judgments
- **Starter code:** Python skeleton with TODOs
- **Evaluation script:** Compute precision, recall, MAP automatically

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Tips

- Start with small subset (100 docs) for debugging
- Use Python dictionaries for inverted index
- Test each component independently before integration
- Compare your TF-IDF values with examples in lecture
- Don't reinvent the wheel: use NLTK for tokenization, stemming

Next Class: Indexing

Preparation

Read:

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What We'll Cover

- How to build indexes for billions of documents
- Block-based indexing algorithms
- Index compression techniques
- Distributed indexing (MapReduce)
- Dense indexes for neural retrieval

Questions?

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Office Hours: [Your time]

Email: [Your email]

Course Website: [Your URL]

Start early on the assignment!

Backup: TF-IDF Variants

Log-Frequency TF

$$\text{tf}(t, d) = \begin{cases} 1 + \log(\text{count}(t, d)) & \text{if count} > 0 \\ 0 & \text{otherwise} \end{cases}$$

- Dampens effect of high term frequencies
- 10 occurrences not 10× more important than 1

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- Normalizes by maximum term frequency in document
- Prevents bias toward long documents

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Probabilistic IDF

$$\text{idf}(t) = \log \frac{N - \text{df}(t) + 0.5}{\text{df}(t) + 0.5}$$

- Used in BM25 (Lecture 3)
- Better theoretical foundation

Backup: Advanced Preprocessing

Named Entity Recognition (NER)

Identify and preserve named entities

- "Barack Obama" $\rightarrow$ single entity
- "New York City" $\rightarrow$ single entity
- Important for proper nouns

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- Language-specific rules

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- "New York City" $\rightarrow$ single entity
- Important for proper nouns

Compound Word Splitting

Handle compound words (especially in German, Dutch)

- "Informationssystem" $\rightarrow$ "Information" + "System"
- Language-specific rules

Case Folding Exceptions

Preserve case for certain terms

- Acronyms: "USA", "IBM"
- Names: "Bush" (president) vs. "bush" (plant)
- Context-dependent decision

Backup: Alternative Similarity Measures

Euclidean Distance

$$\text{dist}(\vec{q}, \vec{d}) = \sqrt{\sum_t (q_t - d_t)^2}$$

- ✗ Sensitive to document length
- ✗ Not normalized
- Rarely used in IR

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Dice Coefficient

$$\text{Dice}(q, d) = \frac{2|q \cap d|}{|q| + |d|}$$

- Similar to Jaccard
- Weighs shared terms more heavily

Backup: Test Collections

Standard IR Test Collections

- **TREC:** Text REtrieval Conference collections (various domains)
- **Cranfield:** Classic small collection (1400 docs, aerodynamics)
- **CACM:** Computer science abstracts (3200 docs)
- **Reuters:** News articles (800,000+ docs)
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Components of Test Collection

1. **Documents:** The corpus to search
2. **Topics/Queries:** Information needs
3. **Relevance judgments (qrels):** Which docs are relevant to which queries
4. Usually created by human assessors (expensive!)